This document is written with the intention of helping the author learn. Not only by teaching, but also in the scenario where a time traveler gifts it to their younger self. Other readers are also welcome, but be warned: this text is written for highly motivated, mathematically mature readers. Readers are expected to be able to supply rigorous proofs where they are omitted, try their hand at proving results before reading the given proofs, internalize the given proofs in their own terms, work out cases on their own, and so on. The author believes the best way to learn is with a mentor. The author knows there are easily available books far better than this document, in every respect, except that it lets the author do as they please. This document, eventually, should be in guided exercise form. Exposition should follow the historical progress and have natural motivation.

Contents

The Fundamental Theorem of Arithmetic

Fact (Principle of Induction)**.** *Any non-empty subset of the positif integers contains a smallest element.*

Fact + Nomenclature. *Let d be a positive integer and m be an integer. Then, there exists a unique pair of integers q, r satisfying $m = dq + r$ and $0 \leq r < d$. We call q the* <u>*quotient*</u> *and r the* <u>*remainder*</u> *of the division of m by d. When $r = 0$, that is, when there exists an integer q with $m = dq$, we say that d* <u>*divides*</u> *m, that d is a* <u>*divisor*</u> *of m, or that m is* <u>*divisible*</u> *by d, and we write this relation as $d \mid m$.*

Exercise. *Compute the multiplication tables modulo n for $n = 7, 8, 10$. Stare at them and look for patterns.*

Nomenclature. *An integer $p \geq 2$ is called a* <u>*prime number*</u> *if its only divisors are 1 and p.*

Fact. *Each positive integer $n \geq 2$ is divisible by at least one prime number p.*

Fact (Primes don't materialize out of thin air)**.** *Let p be a prime number, and let m, n be integers satisfying $p \mid mn$. Then, either $p \mid m$ or $p \mid n$.*

Proof. Without loss of generality, we may restict our attention to positif m, n. By the principle of induction, we may suppose the theorem holds for all prime numbers p' smaller than p (and any n, m). Again by the principle of induction, we may suppose the theorem holds for all positif pairs m', n' with a smaller sum $m' + n' < m + n$ (and our fixed p). Applying division with remainder, we may suppose $0 \leq m, n < p$. We may now write $kp = mn$ for some integer $0 \leq k < p$.
Case 1. If $k = 0$ then either $m = 0$ or $n = 0$, and we have our desired conclusion.
Case 2. We cannot have $k = 1$, by p being a prime number.
Case 3. If $k \geq 2$, then we may pick some prime number q dividing k. In particular $q \mid mn$ and $q < p$. Thus, the (first) induction hypothesis, yields that either $q \mid m$ or $q \mid n$. Without loss of generality, we may assume $q \mid m$. We may write $k = qd$ and $m = qr$ for some positif integers d, r with $r < m$, yielding $dp = rn$. Thus, the (second) induction hypothesis yields that either $p \mid r$ (implying $p \mid m$) or $p \mid n$, reaching our desired conclusion. $\square$

Fact (Fundamental Theorem of Arithmetic)**.** *Each positive integer n may be written as a product of a list of prime numbers $n = p_1 \ldots p_\ell$, in a manner unique up to reordering of the terms.*

Number Theory - Exercises for Lecture 1

Fact. *$d \mid m$ and $m \mid n$ implies $d \mid n$.*

Fact. *m and n have the same remainder with respect to division by d if and only if $d \mid m - n$.*

Fact. *The remainders of $m + n$ and of mn with respect to division by d depends only on the remainders of m and of n with respect to division by d.*

Fact + Nomenclature. *For a positive integer n and a prime number p, let the <u>p-adic valuation</u> $\nu_p(n)$ be the number of times we may divide n by p (until the result is no longer divisible by p). We have*

$$n = \prod_p p^{\nu_p(n)}$$

Fact. *We have $\nu_p(nm) = \nu_p(n) + \nu_p(m)$.*

Fact. *We have $\nu_p(n + m) \geq \min(\nu_p(n), \nu_p(m))$, and equality holds if $\nu_p(n) \neq \nu_p(m)$ (but not necessarily otherwise).*

Fact. *A positive integer is a perfect k-th power if and only if $k \mid \nu_p(n)$ for all prime numbers p. Deduce that the square root of 2 is irrational. More generally, the k-th root of a positive integer which is not a perfect k-th power, is irrational.*

Fact. *The number of divisors of n equals $\prod_p(\nu_p(n) + 1)$. In particular, n has an odd number of divisors if and only if n is a perfect square - convince a child of this.*

Exercise. *Search for Zarmelo's proof of the Fundamental Theorem of Arithmetic, which does not make use of the fact $p \mid mn \implies p \mid m$ or $p \mid n$.*

TODO: fix lecture 1 given the following.

Nomenclature. *We call a invertible modulo n if $0, a, 2a, 3a, \ldots, (n-1)a$ cover all n distinct residue classes modulo n. Namely, if the row corresponding to a in the multiplication table modulo n contains each residue precisely once.*

Fact. *a is invertible modulo n if and only if there exists b such that $ab \equiv 1 \mod n$. In that case, b is unique modulo n.*

Fact. *xy is invertible modulo n if and only if both x and y are invertible modulo n.*

Fact. *Let p be a prime number. Then each non-zero residue a is invertible modulo p.*

Proof. By induction on $a = 1, \ldots, p-1$. For $a = 1$ the result is clear. Assume the result holds for all previous a values. Divide p by a with remainder so that $p = aq + r$ for some $0 \leq r < a$. Since p is a prime number and $1 < a < p$, we have that $a \nmid p$ so $1 \leq r < a$, and thus the inductive hypothesis yields that $r \equiv -aq \mod p$ is invertible modulo p, and hence so is a. $\square$

Nomenclature. *The number of invertible residue classes modulo n is denoted $\varphi(n)$.*

Fact. *The multiplication table modulo n, reduced to the rows and columns of invertible residues modulo n (so that it has size $\varphi(n) \times \varphi(n)$), has only entries that are invertible modulo n, and moreover each row (and column) contains each invertible residues precisely once.*

Fact. *For each invertible residue class a modulo n we have $a^{\varphi(n)} \equiv 1 \mod n$.*

Proof. Let $x_1, \ldots, x_{\varphi(n)}$ list the invertible residue classes modulo n. Then $ax_1, \ldots, ax_{\varphi(n)}$ is a reordering of the same list. Therefore, the products of the two lists are equal, that is $x_1 \ldots x_{\varphi(n)} \equiv ax_1 \ldots ax_{\varphi(n)} = a^{\varphi(n)} x_1 \ldots x_{\varphi(n)} \mod n$. Since $x_1, \ldots, x_{\varphi(n)}$ are invertible modulo n, we have $1 \equiv a^{\varphi(n)} \mod n$ as desired. $\square$

Quadratic Residues and Quadratic Forms

Fact (Thue)**.** *Let a be coprime to n, and $n \geq 2$. Then, there exists a solution to $ax \equiv y \mod n$ with $0 < |x|, |y| < \sqrt{n}$.*

Proof. Consider all combinations $ax - y \mod n$ where $x, y = 0, \ldots, \lfloor\sqrt{n}\rfloor$. There are $(1 + \lfloor\sqrt{n}\rfloor)^2 > n$ such combinations, so by the pigeon hole principle there must be a repetition. So let $ax_1 - y_1 \equiv ax_2 - y_2$ where $(x_1, y_1) \neq (x_2, y_2)$. Since a is coprime to n, we must have that $x_1 \not\equiv x_2$ and $y_1 \not\equiv y_2$. Now $x = x_1 - x_2$ and $y = y_1 - y_2$ satisfy the requested requirments, as desired. □

Fact. *Let $p \equiv 1 \mod 4$. Then, there exists x, y with $p = x^2 + y^2$.*

Proof. Let a solve $a^2 \equiv -1 \mod p$. Applying Thue's result, let x, y solve $ax \equiv y \mod p$ with $0 < |x|, |y| < \sqrt{p}$. Then $0 < x^2 + y^2 < 2p$ and $x^2 + y^2 \equiv x^2(1 + a^2) \equiv 0 \mod p$, so $p = x^2 + y^2$ as desired. □

Proof. Since -1 is a quadratic residue, we may solve $x^2 + y^2 = np$ for some $1 \leq n < p/2$. If $n = 1$ we're done. Otherwise, let X, Y be representatives of x, y modulo n with $|X|, |Y| \leq n/2$. Then we have $X^2 + Y^2 = nk$, for some $1 \leq k \leq n/2$. Now

$$n^2kp = (x^2 + y^2)(X^2 + Y^2) = A^2 + B^2$$

where $A = xX + yY$, $B = xY - yX$ are both divisible by n. Thus

$$kp = \left(\frac{A}{n}\right)^2 + \left(\frac{B}{n}\right)^2$$

giving us descent. □

Algebraic Integers

Fact. *A number which is both rational and an algebraic integer must be an integer.*

$$\mathbf{Q} \cap \overline{\mathbf{Z}} = \mathbf{Z}$$

Fact (Fundamental Theorem of Symmetric Polynomials)**.** *Let $f(x_1, \dots, x_n)$ be a symmetric polynomial. Then, in fact, $f(x_1, \dots, x_n) = g(e_1, \dots, e_n)$ may be given as a polynomial expression in the elementary symmetric polynomials in $x_1, \dots, x_n$.*

Proof. We give a lexicographic ordering of monomials, so that $x_1^{i_1} \dots x_n^{i_n} >_{lex} x_1^{j_1} \dots x_n^{j_n}$ whenever $i_k > j_k$ for the first index k where $i_k \neq j_k$. This is a total order of all monomials. Suppose the lexicographically highest monomial term of $f(x_1, \dots, x_n)$ is $x_I = x_1^{i_1} \dots x_n^{i_n}$, and that the theorem holds for all symmetric polynomials containing terms only lexicographically smaller than this monomial. Since f is symmetric, we have $i_1 \geq \dots \geq i_n$. Now, $E_I = e_1^{i_1 - i_2} e_2^{i_2 - i_3} \dots e_{n-1}^{i_{n-1} - i_n} e_n^{i_n}$ has highest lexicographic monomial term equal to $x_1^{i_1 - i_2} (x_1 x_2)^{i_2 - i_3} \dots (x_1 \dots x_{n-1})^{i_{n-1} - i_n} (x_1 \dots x_n)^{i_n} = x_I$. Subtracting cE_I from f (where c is the coefficient of x_I in f) yields a symmetric polynomial, all of whose terms are lexicographically smaller than x_I, so by induction $f - cE_I$ is expressible as a polynomial in $e_1, \dots, e_n$, and thus so is f, as desired. $\square$

Fact. *Sums and products of algebraic integers are algebraic integers.*

Proof. Let α, β be algebraic integers. Then we may write

$$(x - \alpha_1) \dots (x - \alpha_m) = x^m + a_1 x^{m-1} + \dots + a_{m-1} x + a_m$$

$$(x - \beta_1) \dots (x - \beta_n) = x^n + b_1 x^{n-1} + \dots + b_{n-1} x + b_n$$

for integers a_i, b_j and numbers α_i, β_j with $\alpha = \alpha_1$, $\beta = \beta_1$. Consider the polynomials

$$\prod_{i,j} (x - (\alpha_i + \beta_j))$$

and

$$\prod_{i,j} (x - (\alpha_i \beta_j))$$

By the fundamental theorem of symmetric polynomials, each of these is expressible as a polynomial in x whose coefficients are polynomials in a_i, b_j. That is, these polynomials have integer coefficients. In particular, $\alpha + \beta$ and $\alpha\beta$ are algebraic integers, as desired. $\square$

Proof. Let $A \in \mathbf{Z}^{m \times m}$, $B \in \mathbf{Z}^{n \times n}$ be respectively the companion matrices of α, β. Let A have eigenvalues α_i with corresponding eigenvectors v_i, and let B have eigenvalues β_j with corresponding eigenvectors u_j. Then $A \otimes B \in \mathbf{Z}^{nm \times nm}$ and $A \otimes \mathrm{id}_n + \mathrm{id}_m \otimes B$ have respectively the eigenvalues $\alpha_i \beta_j$ and $\alpha_i + \beta_j$, both with corresponding eigenvectors $v_i \otimes u_j$. It follows that $\alpha_i \beta_j$ and $\alpha_i + \beta_j$ (and in particular $\alpha\beta$ and $\alpha + \beta$) are algebraic integers, as desired. $\square$

Fact. *For $\mathfrak{a} = (\alpha)$ we have $N(\mathfrak{a}) = \left| N_{K/\mathbf{Q}}(\alpha) \right|$.*

Fact. *K has class number 1 iff $\mathcal{O}_K$ is a PID iff $\mathcal{O}_K$ is a UFD.*

Cyclotomics

Nomenclature. *The n-th cyclotomic polynomial is*

$$\Phi_n(x) = \prod_{\mu \text{ a primitive } n\text{-th root of unity}} (x - \mu) = \prod_{j \in (\mathbf{Z}/n\mathbf{Z})^\times} (x - \zeta^j)$$

where $\zeta = e^{2\pi i/n}$.

Fact.

$$x^n - 1 = \prod_{d|n} \Phi_d(x)$$

Fact. $\Phi_n(x) \in \mathbf{Z}[x]$ *is a monic integer polynomial.*

Proof. Clearly $\Phi_n(x)$ is a monic polynomial. By induction, we have that $\Phi_n(x)$ is a rational function of x with rational coefficients, but it has no poles so it is a rational polynomial. $\Phi_n(x) \in \mathbf{Q}(x) \cap \mathbf{C}[x] = \mathbf{Q}[x]$. It is clear that the roots of $\Phi_n(x)$ are algebraic integers, so the coefficients of $\Phi_n(x)$ are algebraic integers that are rational numbers, and so $\Phi_n(x)$ has integer coefficients. $\Phi_n(x) \in \mathbf{Q}[x] \cap \overline{\mathbf{Z}}[x] = \mathbf{Z}[x]$. $\square$

Fact. $\Phi_n(x)$ *is irreducible in* $\mathbf{Q}[x]$.

Proof. (Schur) Recall that the discriminant of $x^n - 1$ is $\pm n^n$. Let $f(x)$ be the minimal polynomial of ζ over $\mathbf{Q}$. So $f(x) \in \mathbf{Z}[x]$ and $f(x) = (x - \zeta_1) \ldots (x - \zeta_k)$ for some distinct n-th roots of unity $\zeta_1, \ldots, \zeta_k$. We must show that the roots of $f(x)$ are closed under exponentiating by powers coprime to n. It suffices to show that if μ is a root of f and p is a prime coprime to n then μ^p is a root of f. We have that $f(\mu^p) = f(\mu^p) - f(\mu)^p = f(x^p) - f(x)^p \restriction_{x=\mu} = pg(x) \restriction_{x=\mu}$ for some $g(x) \in \mathbf{Z}[x]$, so that $p \mid f(\mu^p)$ over $\overline{\mathbf{Z}}$. Moreover, if we assume for the sake of contradiction that μ^p is not a root of f then $f(\mu^p) = (\mu^p - \zeta_1) \ldots (\mu^p - \zeta_k)$ divides the discriminant $\pm n^n$ over $\overline{\mathbf{Z}}$. However, $p \mid \pm n^n$ over $\overline{\mathbf{Z}}$ implies $p \mid n$ over $\mathbf{Z}$, contrary to our assumption. This shows that μ^p is a root of f, as desired. $\square$

Fact. *Let $\zeta = \zeta_{p^r}$, $K = \mathbf{Q}(\zeta)$, and $m = [K : \mathbf{Q}]$. We have*

- $N_{K/\mathbf{Q}}(1 - \zeta) = p$
- $\langle 1 - \zeta \rangle$ *is a prime ideal, lying over p.* $(1 - \zeta)\mathcal{O}_K \cap \mathbf{Z} = p\mathbf{Z}$
- $(1 - \zeta^j)/(1 - \zeta) \in \mathcal{O}_K^\times$ *for all* $j \in (\mathbf{Z}/p^r\mathbf{Z})^\times$
- $p\mathcal{O}_K = (1 - \zeta)^m \mathcal{O}_K$
- $\mathbf{Z}[\zeta] \cap p\mathcal{O}_K = p\mathbf{Z}[\zeta]$
- $\mathcal{O}_K = \mathbf{Z}[\zeta]$

Fermat's Last Theorem

Theorem. *(Wiles, Taylor, Taniyama, Shimura, Ribet, Frey, ..., Fermat) Let $n \geq 3$. Then, any solution in rational numbers to the equation*

$$x^n + y^n = z^n$$

is trivial, in the sense that $xyz = 0$.

Exercise. *It suffices to prove the case $n = 4$ and the cases where $n = p$ is an odd prime. Also, we may assume x, y, z are pairwise relatively prime.*

Exercise. *There are algebraic integers on the unit circle other than the roots of unity.*

However, such numbers have conjugates outside the unit circle.

Fact. *Suppose $f(x) = \prod_\alpha (x - \alpha) \in \mathbf{Z}[x]$ is a monic polynomial with integer coefficients, all of whose roots lie in the closed unit circle. Then each roots α is either zero or a root of unity.*

Proof. Without loss of generality $f(x)$ is irreducible, so it has no repeated roots. Let A be the companion matrix. Then A is diagonable, with eigenvalues all of absolute value at most one. Thus the matrices $(A^n)_{n \geq 1}$ are bounded. However, these are integer matrices. By the pigeonhole principle, we have $A^n = A^{n+k}$ for some $k > 0$. It follows that $\alpha^n = \alpha^{n+k}$, so either $\alpha = 0$ or α is a k-th root of unity. □

Fact. *Any unit u of $\mathbf{Q}(\zeta_p)$ may be written as $u = r\zeta^k$ for some real r and some integer k.*

Fact. *For any $\alpha \in \mathbf{Z}[\zeta_p]$ there is an integer a with $\alpha^p \equiv a \mod \langle 1 - \zeta \rangle^p$.*

Fact (Kummer)**.** *Let p be a regular prime. Then the $n = p$ case of FLT is true.*

Proof. Let us write the equation in the form

$$x^p + y^p + z^p = 0$$

We have

$$\prod_{j=0}^{p-1} (x + \zeta^j y) = x^p + y^p = -z^p$$

And so

$$\prod_{j=0}^{p-1} \langle x + \zeta^j y \rangle = \langle z \rangle^p$$

First let us handle the case $p \nmid xyz$.
Claim. We claim that $\langle x + \zeta^j y \rangle$ are pairwise coprime.
Indeed, suppose $\mathfrak{q} \mid \langle x + \zeta^j y \rangle$, $\mathfrak{q} \mid \langle x + \zeta^k y \rangle$. Then $x + \zeta^j y, x + \zeta^k y \in \mathfrak{q}$. Thus $\zeta^k (1 - \zeta^{k-j}) y \in \mathfrak{q} \implies y \in \mathfrak{q}$ or $1 - \zeta^t \in \mathfrak{q}$. However, if $y \in \mathfrak{q}$ then $x \in \mathfrak{q}$, but this contradicts x, y being coprime. It follows that $1 - \zeta^t \in \mathfrak{q}$. Now $1 - \zeta^t \sim 1 - \zeta$, and $\langle 1 - \zeta \rangle$

is prime, so $\mathfrak{q} = \langle 1-\zeta \rangle$. Thus $z \in \langle 1-\zeta \rangle$. Since $N(\langle 1-\zeta \rangle) = p$, we get $p \mid z$, contradicting our hypothesis.

It follows that each of the ideals $\langle x + \zeta^j y \rangle$ is a p-th power. Now let $\langle x + \zeta y \rangle = \mathfrak{m}^p$. So $\mathfrak{m}$ has order dividing p in the class group. Since p is regular, $\mathfrak{m} = \langle \alpha \rangle$ must be principal. We have

$$x + \zeta y = u\alpha^p$$

for some unit u. We may write $u = r\zeta^k$ for some real r and integer k. We find an integer a with $\alpha^p \equiv a \mod \langle 1-\zeta \rangle^p$ so $\alpha^p \equiv a \mod p$ and hence $x + \zeta y \equiv r\zeta^k a \mod p$. Inverting by ζ^k and conjugating, we find

$$\zeta^{-k}(x + \zeta y) - \zeta^k(x + \zeta^{-1} y) \equiv 0 \mod p$$

So consider the algebraic integer β with $\beta = p^{-1}\left(\zeta^{-k}x + \zeta^{1-k}y - \zeta^k x - \zeta^{k-1}y\right)$. Since $\mathcal{O}_K = \mathbf{Z}[\zeta]$ has integer basis $(1, \zeta, \ldots, \zeta^{p-1}$ minus any particular one), if $\zeta^{-k}, \zeta^{1-k}, \zeta^k, \zeta^{k-1}$ were all distinct we'd get that $p \mid x, y$, contradicting our hypothesis.

Case 1. We have $k \equiv 0 \mod p$. In this case $\langle 1-\zeta \rangle^{p-1} = \langle p \rangle \ni (\zeta - \zeta^- 1)y = \zeta^{-1}(\zeta^2 - 1)y$. Since $\zeta^2 - 1 \sim 1 - \zeta$, we have $y \in \langle 1-\zeta \rangle \cap \mathbf{Z} = p\mathbf{Z}$, contradicting our hypothesis.

Case 2. We have $k \equiv 1 \mod p$. This is the same as the previous case with y replaced by x.

Case 3. We must have $2k \equiv 1 \mod p$. Thus

$$p\beta\zeta^k = (x - y)(1 - \zeta)$$

Taking norms, we have $p \mid x - y$. By symmetry, $x \equiv y \equiv z \mod p$ and so $0 \equiv 3x^p \mod p$. Our assumption yields $p = 3$. However, for $3 \nmid x, y, z$ we have $0 = x^3 + y^3 + z^3 \equiv \pm 1 \pm 1 \pm 1$ mod 9, a contradiction.

TODO handle case $p \mid xyz$ and prove lemmas. $\square$

Gauss Sums and Quadratic Reciprocity

Let p be an odd prime, $\varepsilon_p = (-1)^{\frac{p-1}{2}}$, $p^* = \varepsilon_p p$, $\zeta = e^{2\pi i/p}$. Now, we set

$$\tau = \sum_{x \in \mathbf{F}_p} \zeta^{x^2}$$

We have

$$\tau^* = \sum_{y \in \mathbf{F}_p} \zeta^{-y^2}$$

If $\varepsilon_p = 1$ then -1 is a square modulo p and so $\tau^* = \tau$. Otherwise -1 is a non-square modulo p and so $\tau + \tau^* = 2\sum_{w \in \mathbf{F}_p} \zeta^w = 0$. Thus

$$\tau^* = \varepsilon_p \tau$$

We have

$$\tau\tau^* = \sum_{x,y \in \mathbf{F}_p} \zeta^{x^2-y^2} = \sum_{u,v \in \mathbf{F}_p} \zeta^{uv} = p$$

where we make use of the bijection $(x, y) \longleftrightarrow (u, v)$ given by $u = x + y$, $v = x - y$. It follows that

$$\tau = \pm\sqrt{p^*}$$

Fact. *We have*

$$\tau = \sum_{a \in \mathbf{F}_p} (a : p)\zeta^a$$

Fact. *Let $a_0, \ldots, a_{p-1}$ be integers with $\sum_j a_j \zeta^j = 0$. Then $a_0 = \cdots = a_{p-1}$ are all equal.*

Definition 1. *Let $N_r(a) = \#\{(x_1, \ldots, x_r) \in \mathbf{F}_p^r : x_1^2 + \cdots + x_r^2 = a\}$.*

For example, we have $N_1(a) = 1 + (a : p)$.
Let r be even. Then

$$(p^*)^{r/2} = \tau^r = \sum_{x_1,\ldots,x_r \in \mathbf{F}_p} \zeta^{x_1^2+\cdots+x_r^2} = \sum_{a \in \mathbf{F}_p} N_r(a)\zeta^a$$

So

$$N_r(0) - (p^*)^{r/2} = N_r(a) \quad \forall a \in \mathbf{F}_p^*$$

It follows (since $\sum_a N_r(a) = p^r$) that

$$N_r(a) = \begin{cases} p^{-1}(p^r - (p^*)^{r/2}) & a \neq 0 \\ p^{-1}(p^r + (p-1)(p^*)^{r/2}) & a = 0 \end{cases}$$

Let r be odd. Then

$$(p^*)^{(r-1)/2} \sum_{a \in \mathbf{F}_p} (a : p)\zeta^a = (p^*)^{(r-1)/2}\tau = \tau^r = \sum_{x_1,\ldots,x_r \in \mathbf{F}_p} \zeta^{x_1^2+\cdots+x_r^2} = \sum_{a \in \mathbf{F}_p} N_r(a)\zeta^a$$

So that $N_r(a) - (p^*)^{(r-1)/2}(a : p) \equiv c$ is a constant. It follows that $c = p^{r-1}$ and so

$$N_r(a) = p^{r-1} + (p^*)^{(r-1)/2}(a : p)$$

Fact (Quadratic Reciprocity). *Let p, q be distinct odd primes. Then $(p : q)(q : p) = \varepsilon_p \varepsilon_q$.*

Proof. Let us consider $N_q(1) \mod q$. On one hand, $N_q(1) = p^{q-1} + (p^*)^{(q-1)/2} \equiv 1 + (p^* : q) \mod q$. On the other hand, the sphere $\{(x_1, \ldots, x_q) \in \mathbf{F}_p^q : x_1^2 + \cdots + x_q^2 = 1\}$ is acted on by C_q via cyclic rotations. It follows that $N_q(1) \equiv 1 + (q : p) \mod q$. □

Proof. We have

$$\tau(p^* : q) \equiv \tau(p^*)^{\frac{q-1}{2}} = \tau(\tau^2)^{\frac{q-1}{2}} = \tau^q \equiv \sum_{a \in \mathbf{F}_p} (a : p)\zeta^{aq} = \sum_{a \in \mathbf{F}_p} (aq : p)\zeta^a = (q : p)\tau \mod q$$

Multiply by τ to remove this term. □

Beta Function

Fact.

$$\int_0^1 p^k(1-p)^m = \frac{k!m!}{(k+m+1)!}$$

Proof. Consider the following stochastic process. First, generate $p \sim U[0,1]$. Then, generate n independent Bernoulli(p) trials $b_1, \ldots, b_n \sim \mathrm{Ber}(p)$. What is the probability of having precisely k successes? The answer, of course, is

$$\int_0^1 \binom{n}{k} p^k(1-p)^{n-k} \mathrm{d}p$$

However, instead of generating the b_i as Bernoulli trials out of thin air, let us generate $x_1, \ldots, x_n \in U[0,1]$ and set $b_i = [x_i < p]$. Then, the questions asks what is the probability that precisely k of the n numbers $x_1, \ldots, x_n$ come before p. Clearly, the ordering of $p, x_1, \ldots, x_n$ is uniform over all permutations of $n+1$ terms, and so this probability is $\frac{1}{n+1}$ for each $k = 0, \ldots, n$. $\square$

Finite Fields

Fact. *The field extension* $\mathbf{F}_{q^m}/\mathbf{F}_q$ *is cyclic; The Galois group is generated by the Frobenius automorphism* $x \mapsto x^q$.

Representation Theory

The Classical approach.

Nomenclature. *A* <u>*linear character*</u> *of a finite group G is a homomorphism $\varphi : G \to \mathbf{C}^\times$.*

The set of linear characters will be denoted by $\mathscr{D}(G)$. The trivial character 1 will be denoted by φ_0.

Fact. *A linear character takes values in the unit circle $\mathbf{S}^1$. In fact, it takes values in the group of $|G|$-th roots of unity.*

Fact. *The set of linear characters $\mathscr{D}(G)$ is an abelian group under pointwise multiplication, the unit being the trivial character, and inversion is given by complex conjugation.*

Fact. *If G is cyclic, then $\mathscr{D}(G)$ is cyclic and of the same order.*

Fact. *We have $\mathscr{D}(G \times H) \equiv \mathscr{D}(G) \times \mathscr{D}(H)$.*

Fact. *For abelian groups A we have*

$$\boxed{\mathscr{D}(A) \cong A}$$

(non canonically).

Fact. *In general, we have $\mathscr{D}(G) \equiv \mathscr{D}(G/[G,G])$, and thus $\mathscr{D}(G) \cong G/[G,G]$ (non-canonically).*

Fact. *A linear character φ is always a* <u>*class function*</u>*, meaning $\varphi(hgh^{-1}) = \varphi(g)$.*

Fact. *The only linear characters of S_n are the trivial character and the sign.*

Fact. *We have a homomorphism $\Theta_G : G \to \mathscr{D}(\mathscr{D}(G))$ given by $(\Theta(g))(\phi) = \phi(g)$. If A is abelian, then Θ_A is an isomorphism and $A \equiv \mathscr{D}(\mathscr{D}(A))$.*

Fact (Orthogonality relations)**.** *We have*

$$\sum_{g \in G} \varphi(g) = \begin{cases} |G| & \varphi = \varphi_0 \\ 0 & \varphi \neq \varphi_0 \end{cases}$$

$$\sum_{\varphi \in \mathscr{D}(G)} \varphi(g) = \begin{cases} |\mathscr{D}(G)| & g \in [G,G] \\ 0 & g \notin [G,G] \end{cases}$$

We shall now restict our attention to finite abelian groups A. We write $\hat{A}$ for $\mathscr{D}(A)$.

Fact. *$\hat{A}$ is an orthonormal basis for $\mathbf{C}^A$ under the normalized inner product*

$$\langle f_1, f_2 \rangle = \frac{1}{|A|} \sum_{a \in A} f_1(a) \overline{f_2(a)}$$

Fact + Nomenclature. *For $f : A \longrightarrow \mathbf{C}$ we set $\hat{f} : \hat{A} \longrightarrow \mathbf{C}$ for $\hat{f}(\varphi) = \langle f, \varphi \rangle$. We have that $f \mapsto \hat{f}$ is a linear isomorphism.*

Fact (Plancherel, Parseval)**.** *We have*

$$f = \sum_{\varphi \in \hat{A}} \hat{f}(\varphi)\varphi$$

$$\langle f_1, f_2 \rangle = \sum_{\varphi \in \hat{A}} \hat{f}_1(\varphi)\overline{\hat{f}_2(\varphi)}$$

$$\langle f, f \rangle = \sum_{\varphi \in \hat{A}} \left| \hat{f}(\varphi) \right|^2$$

Fact. *Under the isomorphism $A \equiv \hat{\hat{A}}$, we have*

$$\hat{\hat{f}}(x) = f(x^{-1})$$

When A is additive instead of multiplicative, this reads

$$\hat{\hat{f}}(x) = f(-x)$$

Exercise. *Let $C \in \mathbf{C}^{n \times n}$, $C_{rj} = n^{-1/2} e^{2\pi i r j/n}$. Determine C^2 and C^4.*

Fact. *Let $p(x) = \sum_{m \in \mathbf{Z}} a_m x^m$, where only finitely many terms appear. Then*

$$n^{-1} \sum_{\zeta^n = 1} p(\zeta) = \sum_{n|m} a_m$$

In particular, for $p(x) = \sum_m \binom{N}{m} x^m = (1+x)^N$, we have

$$\sum_{n|m} \binom{N}{m} = n^{-1} \sum_{\zeta^n=1} (1+\zeta)^N$$

Fact (Uncertainty Principle)**.** *Let $f : A \to \mathbf{C}$. Then,*

$$\|f\|_\infty \leq \left\| \hat{f} \right\|_1$$

$$\left\| \hat{f} \right\|_\infty \leq \frac{1}{|G|} \|f\|_1$$

$$f \neq 0 \implies |\mathrm{supp}(f)| \left| \mathrm{supp}(\hat{f}) \right| \geq |G|$$

Fact + Nomenclature (Poisson)**.** *For a subgroup $H \leq A$, we set $H^\perp = \{\varphi \in \hat{A} : \varphi \restriction_H \equiv 1\}$. Then, we have*

1. *$H^\perp$ is a subgroup of $\hat{A}$, and $H^\perp \equiv \widehat{A/H}$.*

2. $\hat{A}/H^{\perp} \equiv \hat{H}$

3. $H^{\perp\perp} \leq \hat{\hat{A}}$ *is isomorphic to* $H \leq A$

4. $$\sum_{\varphi \in H^{\perp}} \varphi(x) = \begin{cases} [A:H] & x \in H \\ 0 & x \notin H \end{cases}$$

5. *For any* $f : A \to \mathbf{C}$ *we have*
$$\sum_{h \in H} f(h) = |H| \sum_{\varphi \in H^{\perp}} \hat{f}(\varphi)$$

Fact (Dedekind)**.** *Let* $f : A \to \mathbf{C}$*, and let* $M \in \mathbf{C}^{A \times A}$ *be given by* $M_{g,h} = f(gh^{-1})$*. Then,*
$$\det M = |A|^{|A|} \prod_{\varphi \in \hat{A}} \hat{f}(\varphi)$$

Fact. *Let* $e_{\varphi^*} = |A|^{-1} \sum_{a \in A} \phi(a)a$*. Then* $(e_{\varphi})_{\varphi \in \hat{A}}$ *are orthogonal idempotents spanning the group algebra* $\mathbf{C}[A]$*.*

Fourier Analysis on the Cube

Fact + Nomenclature (MacWilliams)**.** *For a subspace* $C \leq \mathbf{F}_2^n$ *let*
$$F_C(x, y) := \sum_{v \in C} x^{n - W(v)} y^{W(v)}$$
be the homogeneous generating function of Hamming weights $W(v) = \#\{i \in [n] : v_i = 1\}$ *in* C*. Set* $C^{\perp} = \{v \in F_2^n : \sum v_i c_i = 0 \quad \forall c \in C\}$ *to be the orthogonal space. Then, we have*
$$F_C(x, y) = \frac{1}{|C^{\perp}|} F_{C^{\perp}}(x + y, x - y)$$

Fact. *Let* $f : \mathbf{F}_2^n \to \{\pm 1\}$*. Suppose that* $\Pr(f(x+y) = f(x)f(y) : x, y \sim \mathbf{F}_2^n) \geq 1 - \varepsilon$*. Then, there exists a linear character* $\varphi : \mathbf{F}_2^n \to \{\pm 1\}$ *such that* $\Pr(f(x) = \varphi(x) : x \sim \mathbf{F}_2^n) \geq 1 - \varepsilon$*.*

Proof. Write $f = \sum_{\varphi} \hat{f}(\varphi)\varphi$. We have $I \geq 1 - 2\varepsilon$ for
$I = 2^{-2n} \sum_{x,y \in \mathbf{F}_2^n} f(x+y)f(x)f(y) = 2^{-2n} \sum_{x,y,\varphi_1,\varphi_2,\varphi_3} \hat{f}(\varphi_1)\hat{f}(\varphi_2)\hat{f}(\varphi_3)\varphi_1(x+y)\varphi_2(x)\varphi_3(y)$.
By the orthogonality relations, we need only sum on $\varphi_1 = \varphi_2 = \varphi_3$, so that $I = \sum_{\varphi} \hat{f}(\varphi)^3$.
Thus $1 - 2\varepsilon \leq I \leq \langle f, f \rangle \max \hat{f} = \max \hat{f}$, so there is some linear character φ with
$1 - 2\varepsilon \leq \hat{f}(\varphi)$, i.e. $\Pr(f(x) = \varphi(x) : x \sim \mathbf{F}_2^n) \geq 1 - \varepsilon$, as desired. □

Fact (Arrow)**.** *Consider a voting system which takes as input the preferences of n voters on three candidates, and returns a total preference order on the candidates. Suppose that the system satisfies the following.*

1. *If all voters prefer one candidate over another, the system will reflect so.*
2. *The preference returned by the system on any pair of candidates depends only on the preferences of the voters on that pair of candidates.*
3. *If the voters pick uniform random preferences, so will the system.*

Then, there is a <u>dictator</u>, such that the system always returns their preference.

Proof. Let us call the candidates a, b, c. Let us start with an example with $n = 4$ voters with preferences $a > b > c$, $c > a > b$, $b > a > c$, $b > c > a$. Then the preferences on pairs of candidates are given by $\begin{bmatrix} & a>b>c & c>a>b & b>a>c & b>c>a \\ a:b & a>b & a>b & b>a & b>a \\ b:c & b>c & c>b & b>c & b>c \\ c:a & a>c & c>a & a>c & c>a \end{bmatrix}$ We describe the preferences by three sign vectors $\begin{bmatrix} x & +1 & +1 & -1 & -1 \\ y & +1 & -1 & +1 & +1 \\ z & -1 & +1 & -1 & +1 \end{bmatrix}$. By the second property, the voting system calculates sign functions $f(x), g(y), h(z)$ ranking the candidates, with, for example, $f(x) = 1$ if a is preferred over b and $f(x) = -1$ otherwise. We cannot ever have $f(x) = g(y) = h(z) = \pm 1$. Moreover, out of the 8^n possible sign triples x, y, z, only 6^n represent possible voters preferences. Now we begin our proof. Let us generate a random voter preference (out of the 6^n choices) and describe it using sign vectors x, y, z. We have

$$f(x)g(y) + g(y)h(z) + h(z)f(x) = -1$$

We have

$$\mathbf{E} f(x)g(y) = \sum_{\varphi_1, \varphi_2 \in \widehat{\{\pm 1\}^n}} \hat{f}(\varphi_1)\hat{g}(\varphi_2)\mathbf{E}\varphi_1(x)\varphi_2(y)$$

(but recall how x, y, z is distributed). Now, a linear character on $\{\pm 1\}^n$ is given by $\phi(x) = \prod_{x \in S} x_s$ for some set $S \subseteq [n]$. So we consider $\mathbf{E} \prod_{s \in S} x_s \prod_{t \in T} y_t$. Now, x_i is independent of x_j, y_j whenever $j \neq i$. So if $S \neq T$ then the expected value is zero. Otherwise, the expected value is $(\mathbf{E} x_1 y_1)^{|S|} = \left(\frac{1}{3}(+1) + \frac{2}{3}(-1)\right)^{|S|} = (-1/3)^{|S|}$. We get

$$1 = \sum_{S \subseteq [n]} -(-1/3)^{|S|}(\hat{f}\hat{g} + \hat{g}\hat{h} + \hat{h}\hat{f})(S)$$

By the third property, we have $\hat{f}(\varnothing) = \hat{g}(\varnothing) = \hat{h}(\varnothing) = 0$. Using the bound $|r_1 r_2 + r_2 r_3 + r_3 r_1| \le r_1^2 + r_2^2 + r_3^2$ we get

$$1 \le \sum_{\varnothing \neq S \subseteq [n]} 3^{-|S|} \left(\hat{f}(S)^2 + \hat{g}(S)^2 + \hat{h}(S)^2\right)$$

$$\le 3^{-1} \sum_{\varnothing \neq S \subseteq [n]} \hat{f}(S)^2 + \hat{g}(S)^2 + \hat{h}(S)^2 = 3^{-1}\left(\langle f, f\rangle + \langle g, g\rangle + \langle h, h\rangle\right) = 1$$

This shows that $\hat{f}, \hat{g}, \hat{h}$ are supported on singletons. In other words, f, g, h are linear polynomials. Because they only return two values, they are dictatorial $f(x) = \pm x_k$. By the first property satisfied by the system, $f(x) = x_k$. Lastly, we see that we must have the same dictator. □

Fact (Weierstrass)**.** *Let k be an algebraically closed field, and let A be a commutative k-algebra of finite dimension n over k, with no non-zero nilpotents. Then $A \cong k^n$.*

Proof. By induction on n. The case $n = 1$ is clear. Assume $n > 1$ and let $\zeta \in A \setminus k$. Let $m(x) \in k[x]$ be the minimal polynomial of ζ. Write $m(x) = \prod_j (x - r_j)^{a_j}$ with $r_j \in k$ and $a_j \geq 1$. Then $\prod_j (\zeta - r_j)$ is nilpotent. It follows that $a_j = 1$ for all j. Thus

$$k[\zeta] \cong k[x]/\langle m(x) \rangle \cong \prod_j k[x]/\langle x - r_j \rangle \cong k^d$$

where $d > 1$ is the degree of $m(x)$. It follows that we can find orthogonal idempotents $e_1, \dots, e_d$ spanning $k[\zeta]$. Thus A decomposes as

$$A = e_1 A \oplus \cdots \oplus e_d A$$

and we're done by induction. □

Thus we'd like a criterion ruling out the existence of non-zero nilpotents in our commutative algebra A. Now, if x is nilpotent, then so is xy, and so is the multiplication $m_{xy} \in \mathrm{End}_k(A)$, and thus $\mathrm{Tr}\, m_{xy} = 0$ for all $y \in A$. We notice that $\langle x, y \rangle = \mathrm{Tr}\, m_{x,y}$ is a symmetric bilinear form $A \times A \longrightarrow k$.

Fact. *Let k be field, and let A be a commutative k-algebra of finite dimension n over k. If the form $\langle x, y \rangle = \mathrm{Tr}\, m_{xy}$ is non-degenerate, then A has no non-zero nilpotents. (And the reverse holds in characteristic zero).*

Fact (Frobenius)**.** *Let G be a finite group with s conjugacy classes. Then $Z(\mathbf{C}[G]) \cong \mathbf{C}^s$.*

Proof. By Weierstrass' theorem, we need only show $Z(\mathbf{C}[G])$ has no non-zero nilpotents. By the trace condition, we need only show that $\langle x, y \rangle = \mathrm{Tr}\, m_{xy}$ is non-degenerate. So we shall show that $\det \left(\langle e_i, e_j \rangle \right)_{ij} \neq 0$ for a basis of $\mathbf{Z}(\mathbf{C}[G])$. Let us write $C_1, \dots, C_s$ for the conjugacy classes of G. For a class C_γ, let $e_\gamma = \sum_{g \in C_\gamma} g$. Then $(e_\gamma)_{\gamma=1}^s$ forms a basis of $Z(C[G])$. We have

$$e_k e_j = e_j e_k = \sum_i a_{ijk} e_i$$

where

$$a_{ijk} = \#\{(b, c) \in C_j \times C_k : bc = a\}$$

for any fixed $a \in C_i$. Letting M_γ be the $s \times s$ matrix representing multiplication by e_γ, we have

$$\left(M_\gamma \right)_{\alpha,\beta} = a_{\alpha\beta\gamma}$$

and thus

$$\langle e_\alpha, e_\beta \rangle = \mathrm{Tr}\, M_\alpha M_\beta = \sum_{\gamma,\delta} a_{\gamma\delta\alpha} a_{\delta\gamma\beta}$$

Now, we let

$$h_{ijk} = \#\{(a, b, c) \in C_i \times C_j \times C_k : abc = e\}$$

Since $abc = e \iff bca = e$, we have $h_{ijk} = h_{jki}$. Since $abc = e \iff b(b^{-1}ab)c = e$, we have $h_{ijk} = h_{jik}$. Thus h is symmetric in its variables. For a conjugacy class C_i, we write $C_{i'}$ for the inverse class $\{c^{-1} : c \in C_i\}$. Since $abc = e \iff c^{-1}b^{-1}a^{-1} = e$, we have $h_{ijk} = h_{i'j'k'}$. Let us write

$$h_i = \#C_i$$

so that $h_i = h_{i'}$ and

$$a_{ijk}h_i = h_{i'jk}$$

We are now in position to verify the trace condition $\det\langle e_\alpha, e_\beta\rangle \neq 0$. Permuting the columns, we swap $\beta \leftrightarrow \beta'$. Let

$$P_{\alpha\beta} := \langle e_\alpha, e_{\beta'}\rangle = \sum_{\gamma,\delta} a_{\gamma\delta\alpha}a_{\delta\gamma\beta'} = \sum_{\gamma,\delta} \frac{h_{\gamma'\delta\alpha}h_{\delta'\gamma\beta'}}{h_\gamma h_\delta} = \sum_{\gamma,\delta} \frac{h_{\gamma\delta\alpha}h_{\gamma\delta\beta}}{h_\gamma h_\delta}$$

where in the last step we used $\gamma \leftrightarrow \gamma'$. Let Q be the real $s \times s^2$ matrix

$$Q_{\alpha,(\gamma,\delta)} = \frac{h_{\gamma\delta\alpha}}{\sqrt{h_\gamma h_\delta}}$$

so that

$$P = QQ^t$$

So showing that P is nonsingular reduces to showing that Q has rank s. Indeed, let γ_0 be such that $C_{\gamma_0} = \{e\}$. Then the submatrix $Q_{\alpha,(\gamma_0,\delta)} = \sqrt{h_\delta}[\alpha = \delta']$ is non-singular. $\square$

This allows us to define the irreducible characters of G via the orthogonal idempotents of $Z(\mathbf{C}[G])$, and (I should check) show properties like orthogonality. Only later did Frobenius come up with the notion of a representation.

Exercise. *Find the idempotents for $G = S_3$, the first non-abelian group.*

The Diophantine equation $y^2 = x^3 - 2$

Let $x, y \in \mathbf{Z}$ solve $y^2 = x^3 - 2$. We wish to show that $x = 3, y = \pm 5$. We rewrite

$$x^3 = (y + \sqrt{-2})(y - \sqrt{-2})$$

This suggests working in the ring $\mathbf{Z}[\sqrt{-2}]$. This ring is Euclidean, and so a UFD. Let us show that $y + \sqrt{-2}, y - \sqrt{-2}$ are coprime. Suppose, for the sake of contradiction, that we have $\pi \mid y + \sqrt{-2}$ and $\pi \mid y - \sqrt{-2}$ for some prime element π of $\mathbf{Z}[\sqrt{-2}]$. Then $\pi \mid 2y$, so either $\pi \mid 2$ or $\pi \mid y$. Now $\sqrt{-2}$ is a prime element of $\mathbf{Z}[\sqrt{-2}]$, so in either case $\pi \sim \sqrt{-2} \mid y$. Hence $2 \mid y$. Thus $2 \mid x$. Modulo four, we get $0 \equiv y^2 = x^3 - 2 \equiv 2 \mod 4$, a contradiction. Since $y + \sqrt{-2}, y - \sqrt{-2}$ are coprime, we have

$$y \pm \sqrt{-2} = \varepsilon_\pm z_\pm^3$$

for units ε and some $z \in \mathbf{Z}[\sqrt{-2}]$. The only units are ± 1, which are cubes, so we may remove the ε term. So let us consider

$$y + \sqrt{-2} = (a + b\sqrt{-2})^3 = (a^3 - 6ab^2) + \sqrt{-2}(3a^2 b - 2b^3)$$

We get $1 = 3a^2 b - 2b^3$, so $b = \pm 1$. Also, modulo three, we have $b \equiv 1 \mod 3$. So $b = 1$, and thus $a = \pm 1$. There are two solutions for y, namely ± 5, and $x = 3$.

Exercise. *Solve $y^2 = x^3 - 1$ in integers.*

Sums of Four Squares

Fact. *Every positif integer is a sum of four integer squares.*

Proof. Step 1. Considering that the norm on quaternions is multiplicative, it suffices to show that any odd prime is a sum of four integer squares.
Step 2. By the pigeonhole principle, we may solve $1 + x^2 + y^2 \equiv 0 \mod p$ with $0 \leq x, y < p/2$.
Step 3. Let $n \geq 1$ be minimal such that $np = x^2 + y^2 + z^2 + w^2$ is the sum of four squares. We have $n < \frac{3}{4}p$. We cannot have n even. Indeed, if n were even, wlog x, y have the same parity and so do z, w. Then

$$\frac{n}{2}p = \left(\frac{x+y}{2}\right)^2 + \left(\frac{x-y}{2}\right)^2 + \left(\frac{z+w}{2}\right)^2 + \left(\frac{z-w}{2}\right)^2$$

Step 4. Suppose $n \geq 3$ is odd. Let X, Y, Z, W be the representatives of x, y, z, w modulo n inside $(-n/2, n/2)$. Then $X^2 + Y^2 + Z^2 + W^2 = kn$ for some $0 < k < n$. Now we have

$$n^2 kp = (x^2 + y^2 + z^2 + w^2)(X^2 + Y^2 + Z^2 + W^2) = A^2 + B^2 + C^2 + D^2$$

where $A = xX + yY + zZ + wW$, $B = xY - yX + zW - wZ$, $C = xZ - zX + wY - wY$, $D = yZ - zY + xW - wX$ are all divisible by n. Thus

$$kp = \left(\frac{A}{n}\right)^2 + \left(\frac{B}{n}\right)^2 + \left(\frac{C}{n}\right)^2 + \left(\frac{D}{n}\right)^2$$

is a smaller solution. $\square$

p-adics

Fact. *Let $f \in \mathbf{Z}[x_1, \ldots, x_m]$. Then f is soluble in $\mathbf{Z}_p$ iff f is soluble in $\mathbf{Z}/p^n$ for all n.*

Fact. *Units in $\mathbf{Z}_p$ are precisely those not divisible by p.*

Fact. *Rationality $\iff$ eventual periodicity of p-adic expansion.*

Fact. *The non-zero ideals of $\mathbf{Z}_p$ are precisely $p^n\mathbf{Z}_p$ and $\mathbf{Z}_p/p^n\mathbf{Z}_p \equiv \mathbf{Z}/p^n\mathbf{Z}$.*

The Diophantine equation $x^2+7=2^n$

Let x, n be integers with $x^2+7=2^n$. We claim that
$(x,n)=(\pm 1,3),(\pm 3,4),(\pm 5,5),(\pm 11,7),(\pm 181,15)$.
First, if $n=2k$ is even, then $7=(2^k+x)(2^k-x)$, so $2^k \pm x=7$ and $2^k \mp x=1$, yielding The solution $(\pm 3,4)$. In general, we have $(x+\sqrt{-7})(x-\sqrt{-7})=2^n$. Suggesting to work in the ring of integers of $\mathbf{Q}(\sqrt{-7})$. This is Euclidean, so a UFD. Now $\sqrt{-7}, \dfrac{1 \pm \sqrt{-7}}{2}$ are non-associate irreducible elements, ± 1 are the only units, and $2=\dfrac{1+\sqrt{-7}}{2}\dfrac{1-\sqrt{-7}}{2}$. We may assume $n \geq 5$ is odd, and of course x is odd. We have
$\dfrac{x+\sqrt{-7}}{2} \cdot \dfrac{x-\sqrt{-7}}{2}=\left(\dfrac{1+\sqrt{-7}}{2}\right)^m\left(\dfrac{1-\sqrt{-7}}{2}\right)^m$, where $m=n-2$. We see that
$\dfrac{x \pm \sqrt{7}}{2}$ must be coprime and conjugate, so either $\dfrac{x+\sqrt{-7}}{2}=\pm\alpha^m$ and $\dfrac{x-\sqrt{-7}}{2}=\pm\beta^m$ or $\dfrac{x+\sqrt{-7}}{2}=\pm\beta^m$ and $\dfrac{x-\sqrt{-7}}{2}=\pm\alpha^m$. Thus $\alpha^m-\beta^m=\pm\sqrt{-7}=\pm(\alpha-\beta)$. We claim we cannot have $+$. Indeed, we have $\alpha+\beta=1$ and $\alpha\beta=2$. Thus $\alpha^2 \equiv 1 \mod \beta^2$. Thus $\alpha-\beta=\alpha^m-\beta^m \equiv \alpha \mod \beta^2$, nonsense. So $-\sqrt{-7}=\alpha^m-\beta^m$. Using the binomial expansion, we get $-2^{m-1} \equiv m \mod 7$. Since m is also even, and the multiplicative order of 2 modulo 7 is three, we have that $m \equiv 3,5,13$ modulo 42. We have solutions $m=3,5,13$. Suppose we have two solutions m_1, m_2 equivalent modulo 42, with $m_2>m_1$. Write
$k=m_2-m_1=6 \cdot 7^l s$ where $7 \nmid s$. Now, we have

$$(1+\sqrt{-7})^k \equiv 1+k\sqrt{-7} \mod 7^{l+1}$$

$$2^k \equiv 1 \mod 7^{l+1}$$

So we get $k(\alpha^{m_1}+\beta^{m_1})\sqrt{-7} \equiv 0 \mod 7^{l+1}$, which is easily a contradiction.

Minkowski's Theorem

Fact. *Let E be an n-dimensional Euclidean space, Γ be a full lattice in E, and X a centrally symmetric, convex subset of E. If moreover*

$$\mathrm{vol}(X) > 2^n \,\mathrm{vol}(\Gamma)$$

then X contains a non-zero point of Γ.

Proof. Suppose not. Let P be a fundamental parallelepiped of 2Γ. The map $\Phi : E \longrightarrow E/(2\Gamma) \cong P$ is injective on X. Then $X = \bigsqcup_{\gamma \in 2\Gamma} X_\gamma$ where $X_\gamma = (X \cap (P + \gamma))$. Thus $\Phi(X) = \bigsqcup_{\gamma \in 2\Gamma} \Phi(X_\gamma)$. However, $\mathrm{vol}(\Phi(X_\gamma)) = \mathrm{vol}(X_\gamma)$, so $\mathrm{vol}(X) = \mathrm{vol}(\Phi(X)) \le \mathrm{vol}(P) = \mathrm{vol}(\Gamma)/2^n$. □

Fact. *Let K be a number field of dimension n, and $\Phi : K \to \mathbf{R}^{r_1} \times \mathbf{C}^{r_2} \equiv R^n$ be its embeddings. Then $\Phi(\mathcal{O}_K)$ is a lattice of volume $2^{-r_2}\sqrt{\mathrm{disc}_K}$. More generally, $\Phi(\mathfrak{a})$ is a lattice of volume $2^{-r_2}\sqrt{\mathrm{disc}_K}N(\mathfrak{a})$.*

Fact. *Let K be a number field, $\mathfrak{a}$ a non-zero ideal in $\mathcal{O}_K$. Then there is some $x \in \mathfrak{a}$, $x \neq 0$, with $\left|N_{K/\mathbf{Q}}(x)\right| \le M_K N(\mathfrak{a})$.*

Proof. (Sketch). Work with the set $B_r = \{(x_1, \ldots, x_{r_1}, z_1, \ldots, z_{r_2}) \in \mathbf{R}^{r_1} \times \mathbf{C}^{r_2} : \sum |x_i| + 2\sum |z_j| \le r\}$. □

Unique Factorization to Prime Ideals

Fact. *Let K be a number field. Then any non-zero ideal $\mathfrak{a} \lhd \mathcal{O}_K$ may be written as a product of prime ideals*

$$\mathfrak{a} = \mathfrak{p}_1 \dots \mathfrak{p}_k$$

in a unique manner (up to reordering).

Fact. *Let K be a number field, $\mathfrak{a}, \mathfrak{b}, \mathfrak{c}$ ideals of $\mathcal{O}_K$. If $\mathfrak{a}$ is non-zero, then*

$$\mathfrak{a}\mathfrak{b} = \mathfrak{a}\mathfrak{c} \implies \mathfrak{b} = \mathfrak{c}$$

Fact. *Let K be a number field, $\mathfrak{a}, \mathfrak{b}$ ideals of $\mathcal{O}_K$. Then*

$$\mathfrak{a} \subseteq \mathfrak{b} \iff \mathfrak{b} \mid \mathfrak{a}$$

Fact. *Let K be a number field, $\mathfrak{a}, \mathfrak{b}, \mathfrak{p}$ be ideals of $\mathcal{O}_K$, where $\mathfrak{p}$ is prime. Then*

$$\mathfrak{p} \mid \mathfrak{a}\mathfrak{b} \implies \mathfrak{p} \mid \mathfrak{a} \textit{ or } \mathfrak{p} \mid \mathfrak{b}$$

Fact. *Let K be a number field. Then any ideal $\mathfrak{a}$ of $\mathcal{O}_K$ is divisible by some prime ideal $\mathfrak{p}$ of $\mathcal{O}_K$.*

Fact. *Let K be a number field. Then there is no infinite chain of ideals $\mathfrak{a}_1 \subsetneq \mathfrak{a}_2 \subsetneq \mathfrak{a}_3 \subsetneq \dots$ of $\mathcal{O}_K$.*

Fact. *Let K be a number field, $\mathfrak{a}$ a non-zero ideal of $\mathcal{O}_K$. Then there exists a non-zero ideal $\mathfrak{b}$ of $\mathcal{O}_K$ such that $\mathfrak{a}\mathfrak{b}$ is a principal ideal.*

Absolute Values

Nomenclature. *Let K be a field. An absolute value on K is a function $|\cdot| : K \longrightarrow \mathbf{R}_{\geq 0}$ satisfying the properties*

1. $|x| = 0 \iff x = 0$
2. $|xy| = |x|\,|y|$
3. $|x+y| \leq |x| + |y|$

for all $x, y \in K$.
If $|x+y| \leq \max(|x|, |y|)$ for all $x, y \in K$, we call $|\cdot|$ non-Archimedean.

We have $|\pm 1| = 1$ (and in fact $|\zeta| = 1$ for any root of unity ζ). We have $|x^{-1}| = |x|^{-1}$. An absolute value equips K with the metric $d_{|\cdot|}(x,y) = |x-y|$. The trivial absolute value is $|x| = [x \neq 0]$. Raising an absolute value to a power $\lambda \in (0,1]$ yields an absolute value. Raising a non-Archimedean absolute value to a power $\lambda \in (0,\infty)$ yields a non-Archimedean absolute value.

Fact. *Two absolute values $|\cdot|_1, |\cdot|_2$ enrich K with the same topology if and only if there exists some $\lambda > 0$ with $|x|_2 = |x|_1^\lambda$ for all $x \in K$.*

Proof. The if direction is clear, so we show the only if direction. The set $\{x : |x|_1 < 1\}$ is the set $\{x : \lim_n x^n = 0\}$, so it is determined by the topology. Thus the topology also determines the inverse set $\{x : |x|_1 > 1\}$, as well as $\{x : |x| = 1\}$. We may assume $|\cdot|$ is non-trivial, and fix y with $a = |y|_1 > 1$ and $b = |y|_2 > 1$. Let $x \in K^\times$, and write $|x|_1 = a^\alpha$ for some $\alpha \in \mathbf{R}$. If $q = m/n$ is a rational number with $q < \alpha$, then $|x|_1 > a^q \implies |x^n|_1 > a^m = |y^m|_1 \implies |x^n y^{-m}|_1 > 1 \implies |x^n y^{-m}|_2 > 1 \implies |x|_2 > b^q$. Similarly, we have that $|x|_2 < b^q$ for all rational numbers q with $q > \alpha$. Thus $|x|_2 = b^\alpha = |x|_1^\lambda$ where $\lambda = \log b / \log a$. □

Fact. *Let $|\cdot|$ be a non-Archimedean absolute value. Then we have $|x| \neq |y| \implies |x+y| = \max(|x|, |y|)$.*

Fact. *Let $|\cdot|$ be an absolute value on K. Then $|\cdot|$ is non-Archimedean if and only if the integers are bounded (meaning $|n| \leq C$ for all integers n and some constant C).*

Proof. The only if direction is clear, so we show the if direction. Let $x, y \in K$. Then

$$|x+y|^n = |(x+y)^n| = \left|\sum_k \binom{n}{k} x^k y^{n-k}\right| \leq \sum_k \left|\binom{n}{k}\right| |x|^k |y|^{n-k} \leq (n+1)C\max(|x|, |y|)^n$$

taking n-th root and a limit we get $|x+y| \leq \max(|x|, |y|)$. □

Nomenclature. *Let K be a field, let $|\cdot|$ be an absolute value on K, and let V be a vector space over K. A norm on V is a function $\|\cdot\| : V \to \mathbf{R}_{\geq 0}$ satisfying the properties*

1. $\|v\| = 0 \iff v = 0$

2. $\|\alpha v\| = |\alpha| \, \|v\|$

3. $\|u + v\| \le \|u\| + \|v\|$

for all vectors $u, v \in V$ and scalars $\alpha \in K$.

A pair of norms $\|\cdot\|_1, \|\cdot\|_2$ on V are called equivalent if there are constants $c, C > 0$ such that

$$c \|v\|_2 \le \|v\|_1 \le C \|v\|_2$$

for all $v \in V$.

Fact. *Assume K is complete and V is finite dimensional. Then, any two norms on V are equivalent, and any norm is complete.*

Proof. By induction on $d = \dim V$. The cases $d = 0, 1$ are clear, so let $d \ge 2$. Let $\|\cdot\|$ be any norm on V. Let us fix a basis $e_1, \ldots, e_d$ of V, and define

$$\|v\|_\infty = \max(|x_i|)$$

where $v = x_1 e_1 + \cdots + x_d e_d$. Then $\|\cdot\|_\infty$ is indeed a norm on V, and it will suffice to show that $\|\cdot\|, \|\cdot\|_\infty$ are equivalent, and that $\|\cdot\|_\infty$ is complete. For completeness, we note that a sequence of vectors is Cauchy under $\|\cdot\|_\infty$ if and only if each of the d coordinates is Cauchy, and the same goes for convergence. So let us show that $\|\cdot\|, \|\cdot\|_\infty$ are equivalent. We have one inequality

$$\|v\| \le (\|e_1\| + \cdots + \|e_n\|) \|v\|_\infty$$

for all $v \in V$. Suppose, for the sake of contradiction, that we do not have a constant $c > 0$ with

$$c \|v\|_\infty \le \|v\|$$

for all $v \in V$. Then, we may find a sequence of vectors $(v_n)_n$ with $\|v_n\|_\infty = 1$ and $\|v_n\| \longrightarrow 0$. Restricting to a subsequence, we may assume that the d-th coordinate of each v_n has absolute value 1. Dividing by those coordinates, we may assume that the d-th coordinate of each v_n is 1. Write $v_n = e_d + u_n$ where $u_n \in \operatorname{span}(e_1, \ldots, e_{d-1})$. We get that u_n converges to $-e_n$ under $\|\cdot\|$. However, by induction, the resticted norm $\|\cdot\|$ on $\operatorname{span}(e_1, \ldots, e_{d-1})$ is complete, and so $\operatorname{span}(e_1, \ldots, e_{d-1})$ is closed in V under $\|\cdot\|$. This implies $-e_n \in \operatorname{span}(e_1, \ldots, e_{d-1})$, a contradiction. □

Fact. *Let $|\cdot|$ be a non-Archimedean absolute value on K. Then any point in any ball is the center of the ball.*

Fact. *Any absolute value on $\mathbf{Q}$ is either trivial, equivalent to the usual absolute value, or equivalent to the p-adic absolute value.*

Fact. *Let K be a complete field under an Archimedean absolute value $|\cdot|$. Then $K, |\cdot|$ is isomorphic to either $\mathbf{R}$ or $\mathbf{C}$ with the usual absolute value.*

Proof. (Sketch). Suppose K extends $\mathbf{C}$ with the usual absolute value, and $K \neq \mathbf{C}$. Let $z \in K \setminus \mathbf{C}$. We may find $\lambda \in \mathbf{C}$ closest to z, using compactness of closed balls in $\mathbf{C}$. Without loss of generality, we may assume $\lambda = 0$. Set $m = |z|$. So $m > 0$ and $|z - \alpha| \geq m$ for all $\alpha \in \mathbf{C}$. Now suppose $|\alpha| < m$. Then $z^n - \alpha^n = \prod(z - \zeta^i \alpha)$ so
$|z - \alpha|\, m^{n-1} \leq m^n + |\alpha|^n$ so $|z - \alpha| = m$. This clearly means $|z - \alpha| = m$ for all $\alpha \in \mathbf{C}$, making $|\cdot|$ bounded on $\mathbf{C}$. $\square$

Fact. *Let K be a complete field under the absolute value $|\cdot|$. Suppose E/K is a finite field extension. Then there is exactly one extension of the absolute value $|\cdot|$ to an absolute value of K.*

Galois Theory

Fact. *Let G be a group, and let F be a field. Then, the set of group homomorphisms $G \longrightarrow F^\times$ is linearly independent over F.*

Proof. Suppose, for the sake of contradiction, that $\chi_1, \ldots, \chi_d : G \longrightarrow F^\times$ are distinct homomorphisms that are linearly dependent. We may further assume that d is minimal. Let $a_1, \ldots, a_d$ be scalars, not all zero, with

$$a_1\chi_1 + \cdots + a_d\chi_d = 0$$

By the minimality of d, we have that all the a_i are non-zero. Now, setting $g \leftarrow hg$ we have

$$a_1\chi_1(h)\chi_1 + \cdots + a_d\chi_d(h)\chi_d = 0$$

for each $h \in G$. By the minimality of d, the two equations must be scalar multiples of one another. Thus $\chi_1(h) = \cdots = \chi_d(h)$. Since h is arbitrary, $\chi_1 = \cdots = \chi_d$. So $d = 1$ and $\chi_1 = 0$, which is nonsense. □

Nomenclature. *We set* $\mathrm{Aut}(L/K) := \{\varphi : L \longrightarrow L$ *is a field isomorphism,* $\varphi \restriction_K = \mathrm{id}_K\}$.

Fact. *Let L/K be a finite and separable field extension. Then $L = K(\alpha)$ for some $\alpha \in L$.*

Proof. If K is finite then so is L, so we may choose α to generate the cyclic group $L^\times$. So we assume K is infinite. Now, because L/K is finite and separable, the number of subfields $L/F/K$ is finite. Suppose we have a simple extension $K(\alpha) \neq L$. Pick $\beta \in L \setminus K(\alpha)$. By the pigeonhole principle, the simple extensions $K(\alpha + t\beta)$ ranging over $t \in K$ must have a repetition. If $F = K(\alpha + t_1\beta) = K(\alpha + t_2\beta)$ with $t_1, t_2 \in K$ distinct, then in particular $\alpha, \beta \in K$, so F is a simple extension larger than $K(\alpha)$. But this can't go on forever. □

Fact. *Suppose L/K and L'/K are Galois. Then so is LL'/K.*

Fact. *Suppose L/K is Galois, and L'/K is finite. Then $[LL' : L'] = [L : L \cap L']$.*

Fact. *Suppose two finite field extensions L/K and L'/K (inside some L^{alg}) have $L \cap L' = K$, and at least one is Galois. Then $[LL' : K] = [L : K][L' : K]$.*

Commutative Banach Algebras

Fact. *Let X be a compact Hausdorff space. Then*

$$\boxed{\operatorname{mSpec}\mathscr{C}(X) \equiv X}$$

Proof. For each $x \in X$, $\mathrm{ev}_x : X \to \mathbf{C}$, $\mathrm{ev}_x : f \mapsto f(x)$ is a homomorphism onto a field, so the kernel $\mathfrak{m}_x = \{f \in \mathscr{C}(X) : f(x) = 0\}$ is a maximal ideal. By normality, the $\mathfrak{m}_x$ are distinct. Now let $\mathfrak{m}$ be a maximal ideal. Suppose, for the sake of contradiction, that $\mathfrak{m} \neq \mathfrak{m}_x$ for each $x \in X$. Then we may pick $f_x \in \mathfrak{m}$ with $f_x(x) \neq 0$. Let U_x be a neighbourhood of x where f_x does not vanish. Find a finite cover $X = U_{x_1} \cup \cdots \cup U_{x_n}$. Then $f = f_{x_1}\overline{f_{x_1}} + \cdots + f_{x_n}\overline{f_{x_n}}$ satisfies that $f \in \mathfrak{m}$ and $f > 0$, so f is invertible and $\mathfrak{m} = \langle 1 \rangle$, a contradiction. $\square$

Zeta Functions

Exercise. *Find formulas for* ζ_K *where* $K = \mathbf{Q}(i)$*. Deduce that* $r_2(n) = \sum_{d|n} \chi(d)$ *where* χ *is the non-trivial multiplicative character modulo four. Deduce the Gregory series.*

Exercise. *Find formulas for* ζ_K *where* $K = \mathbf{Q}(\sqrt{-2})$*. Deduce that* $\sum_n \chi(n)/n = \pi/2^{3/2}$ *where* $\chi(n) = 1$ *for* $n \equiv 1, 3 \mod 8$ *and* $\chi(n) = -1$ *for* $n \equiv 5, 7 \mod 8$*.*

Exercise. *Apply Minkowski's theorem to show the following. If* $(-5 : p) = 1$*, then either* $p = x^2 + 5y^2$ *or* $2p = x^2 + 5y^2$*.*

Euclidean domains, PIDs, UFDs, Dedekind Domains

Nomenclature. *A domain A is* Euclidean *if there is a function $\phi : A \setminus \{0\} \to \mathbf{Z}_{\geq 0}$ such that for all $a, b \in A$ with b non-zero, there exists q, r in A with*

$$a = bq + r$$

and with

$$r = 0 \text{ or } \phi(r) < \phi(b)$$

Nomenclature. *A* Principal Ideal Domain *is a domain A in which all ideals are principal.*

Fact. *A Euclidean domain is a PID.*

Nomenclature. *A* Unique Factorization Domain *is a domain A such that for each non-zero, non-unit $a \in A$, there exists a factorization*

$$a = p_1 \dots p_k$$

where the p_i are irreducible, and moreover where such factorizations are unique up to reordering and associates.

Fact. *A PID is a UFD.*

Proof. Let A be a PID, so A is Noetherian, so we cannot have a strictly increasing infinite sequence of ideals of A. This implies that any non-zero, non-unit element a is given as a product of irreducibles. Now it remains to show that if p is irreducible then it is prime. Indeed, $\langle p \rangle$ is maximal among principal ideals, so it is a maximal ideal, hence a prime ideal, so p is prime. □

Fact. *In a UFD, an element is irreducible iff it is prime.*

Fact. *If A is a UFD then so is $A[x]$.*

An example of a UFD which is not a PID is $\mathbf{Z}[x]$. The ideal $\langle 2, x \rangle$ is not principal.
Let us axiomatize the theory of unique factorization into prime ideals in the ring of integers of a number field.

Nomenclature. *A* Dedekind domain *is a Noetherian, integrally closed domain, in which every non-zero prime ideal is maximal.*

Fact. *In a Dedekind domain, there is unique factorization of non-zero ideals into prime ideals.*

Fact. *Let A be a Dedekind domain. Then any ideal $\mathfrak{a}$ of A is generated by at most two elements. In fact, for any non-zero element α in $\mathfrak{a}$ there is a complementary generator.*

Proof. Let $\mathfrak{a} \lhd A$ be a given non-zero ideal, and write $\mathfrak{a} = \mathfrak{p}_1^{e_1} \dots \mathfrak{p}_n^{e_n}$. Given any non-zero $\alpha \in \mathfrak{a}$, we write $(\alpha) = \mathfrak{p}_1^{f_1} \dots p_n^{f_n} \mathfrak{q}_1^{g_1} \dots \mathfrak{q}_m^{g_m}$. We have $(\alpha) \subseteq \mathfrak{a}$ so $\mathfrak{a} \mid (\alpha)$ so $f_i \geq e_i$. Now suppose we've found some $\beta \in A$ with $\nu_{\mathfrak{p}_i}(\beta) = e_i$ and $\nu_{\mathfrak{q}_j}(\beta) = 0$. Then we'd have $(\alpha, \beta) = \gcd((\alpha), (\beta)) = \mathfrak{p}_1^{e_1} \dots \mathfrak{p}_n^{e_n} = \mathfrak{a}$. So it remains to find such β. To do so, pick some $\gamma_i \in \mathfrak{p}_i^{e_i} \setminus \mathfrak{p}_i^{e_i+1}$. By the chinese remainder theorem, we may solve $\beta \equiv \gamma_i \mod \mathfrak{p}_i^{e_i+1}$, $\beta \equiv 1 \mod \mathfrak{q}_j$. □

Fact. *Let A be a Dedekind domain with field of fractions K, let L/K be a finite field extension. Then the integral closure B of A in L is a Dedekind domain, and its field of fractions is L.*

Fact. *If L/K is Galois, then* $\mathrm{Gal}(L/K)$ *acts transitively on the primes $\mathfrak{P}$ over $\mathfrak{p}$.*